RAJALAKSHMI ENGINEERING COLLEGE

25 years OF ACADEMIC EXCELLENCE

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

T075 - Voice-Based AI Receptionist Agent for Smart College Communication - Second Review PPT

SDGs Covered

SDG 04 - Quality Education

SDG 09 – Industry, Innovation and Infrastructure

SDG 16 – Peace, Justice and Strong Institutions

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INTRODUCTION

DOMAIN: Voice AI & Artificial Intelligence

TECHNOLOGIES : TTS, STT, RAG & LLM

- Colleges handle a large number of routine enquiries related to **admissions, courses, fees, facilities, placements, and academic information**.
- Manual telephone-based enquiry handling can lead to **receptionist workload, waiting time, and inconsistent information access**.
- A **Voice-Based AI Receptionist** provides an automated conversational interface for handling institutional information enquiries.
- The system accepts **spoken queries**, understands the request, retrieves relevant college information, and generates a suitable response..
- The implemented system connects **voice communication and institutional knowledge** into a single automated enquiry process.

IMPORTANCE OF THE TOPIC

- Reduces the dependency on manual handling of **routine information enquiries**.
- Makes institutional information **easier to access through natural voice interaction**.
- Delivers **consistent, knowledge-based responses**.
- Helps provide **consistent responses based on official college information**.
- Demonstrates the practical integration of **Voice AI with institutional knowledge retrieval**.

PROJECT OVERVIEW & OBJECTIVES/SCOPE

PROJECT OVERVIEW

- Develops a **Voice-Based AI Receptionist** for automated college information enquiries.
- Enables callers to interact with the system through a **telephone-based voice interface**.
- Processes spoken enquiries and converts them into **text-based queries for information retrieval and response generation**.
- Uses an **institution-specific knowledge base** created from official college webpages and academic documents.
- Generates responses using **retrieved institutional information and a local language model**.
- Provides the generated information back to the caller as **natural speech**, supporting multi-turn interaction.

KEY OBJECTIVES

- Automate **routine institutional information enquiries** through voice interaction.
- Build a reliable **college-specific knowledge base** from official information sources.
- Retrieve relevant information before generating responses to improve **answer grounding**.
- Enable **multi-turn voice conversations** for follow-up enquiries.
- Evaluate the system for **retrieval accuracy, answer correctness, refusal handling, and processing latency**.

SCOPE

- The project covers **voice-based college enquiries, institutional knowledge retrieval, LLM response generation, and speech-based responses** using official college information sources.

PROBLEM STATEMENT

Educational institutions receive a high volume of telephone enquiries about admissions, departments, fees, facilities, placements, and academic information. However, manual handling can lead to delays, inconsistent responses, and limited availability, creating a need for an automated voice-based information system.

KEY PROBLEMS

High Volume of Routine Inquiries – Reception staff repeatedly handle similar queries.

Limited Availability – Assistance is restricted to working hours and staff availability.

Delayed Responses – Callers may experience waiting time during busy periods.

Inconsistent Information – Responses may vary depending on the staff member or available information.

Difficulty Handling Information Queries – Finding relevant information manually can be time-consuming.

WHY IS IT IMPORTANT?

- Reduces the workload of reception staff.
- Provides **faster and consistent information** to callers.
- Improves accessibility to **institutional information through voice interaction**.
- Reduces dependency on manual information lookup.
- Enables **automated handling of routine enquiries**.

EXISTING SYSTEMS VS OUR'S

Factor	Existing Systems	Our System – Voice AI Receptionist
Call Handling	Human receptionists / AI receptionist products (e.g., Smith.ai, GoodCall, My AI Front Desk, PolyAI)	✓ AI-based telephone enquiry handling
Availability	Staff working hours / commercial AI services	✓ Automated voice access (24×7)
Information Access	Websites, documents, FAQs, or configured business knowledge	✓ Institution-specific RAG knowledge base
Interaction	Manual, IVR, chatbot, or AI voice interaction	✓ Natural voice conversation
Knowledge Grounding	Depends on configured knowledge/source integration	✓ Retrieved institutional context before response
Domain Focus	General business / customer-service use cases	✓ College-specific institutional enquiries
Scalability	Usage limits or platform-dependent scalability	✓ Easily extendable with updated college documents
Integration with College Systems	Limited or requires custom integrations	✓ Designed to integrate with official college information sources

WHY THIS PROJECT INSTEAD OF EXISTING PRODUCTS?

- Commercial AI receptionist products are mainly designed for general business and customer-service use cases. This project specifically focuses on **college-related information enquiries**, using **official academic and regulatory information** as its knowledge source. It provides a domain-specific voice interface designed around the information structure and enquiry requirements of an educational institution.

REALTIME CHALLENGES FACED

- **Speech Recognition**
  Handling accents and **Tamil-English code-mixed speech**.
- **Low-Latency Processing**
  Managing the STT → RAG → LLM → TTS processing chain..
- **Accurate RAG Retrieval**
  Retrieving the correct information from **website and PDF content**.
- **Response Grounding**
  Preventing hallucinations and unsupported responses.
- **Multi-Turn Conversation**
  Maintaining context across follow-up queries..
- **Telephony Integration**
  Maintaining reliable bidirectional audio through Asterisk and AudioSocket.

METHODOLOGY

VOICE AI PROCESSING PIPELINE

1. CALL & AUDIO CAPTURE

- Caller connects through **MicroSIP → Asterisk**.
- Asterisk sends bidirectional 8-kHz audio through **AudioSocket**.

2. SPEECH DETECTION & STT

- **WebRTC VAD** detects the beginning and end of speech.
- **Faster-Whisper (Small, CUDA/FP16)** converts the caller's speech into text.

3. QUERY PROCESSING & RAG

- The transcribed query is processed by the **RAG service**.
- **Sentence-Transformers** generates embeddings.
- **ChromaDB** retrieves the most relevant institutional information.
- Retrieved context is provided **to Qwen3:4B through Ollama**.

4. GROUNDED RESPONSE GENERATION

- The local LLM generates a response using the retrieved college context.
- If sufficient information is unavailable, the **system refuses instead of generating an unsupported answer**.

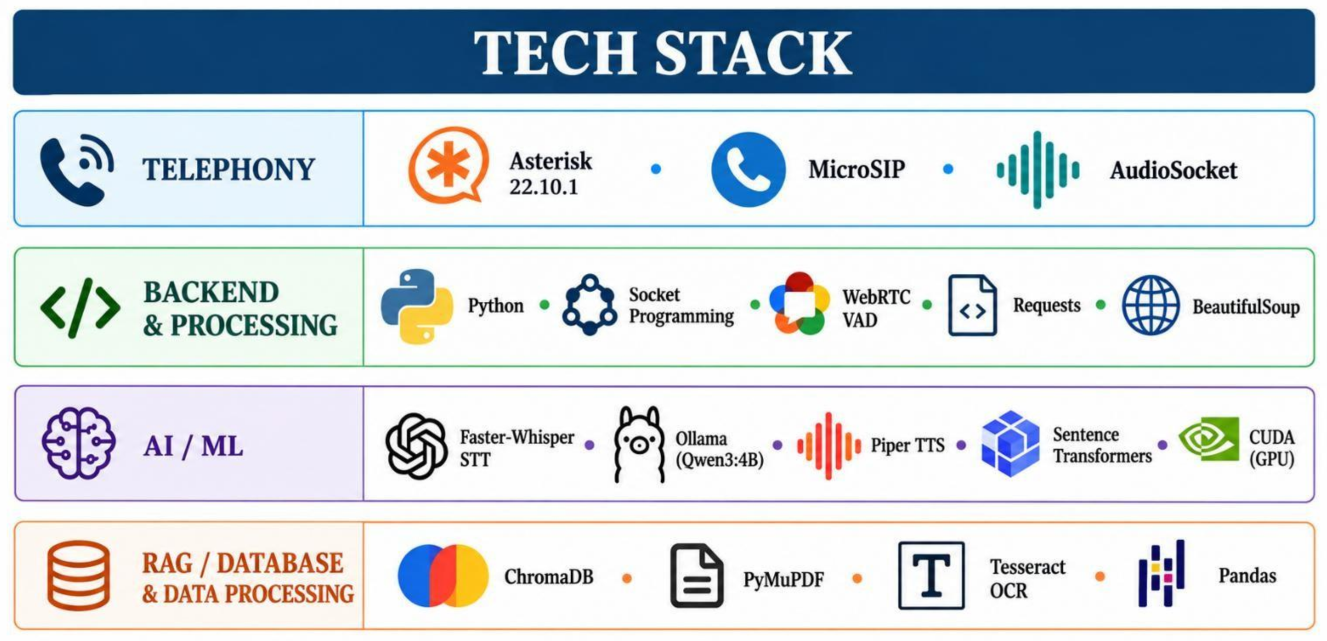
5. VOICE RESPONSE

- **Piper TTS** converts the generated response into speech.
- Audio is converted to **16-bit, 8-kHz mono PCM** and sent through AudioSocket.
- Caller receives the response through MicroSIP.

OVERALL FLOW

Caller Speech → VAD → Faster-Whisper STT → RAG Retrieval → Qwen3:4B → Piper TTS → AudioSocket → Caller→ Escalation identified if required

TECH STACK



DATA SOURCES

COLLEGE KNOWLEDGE BASE

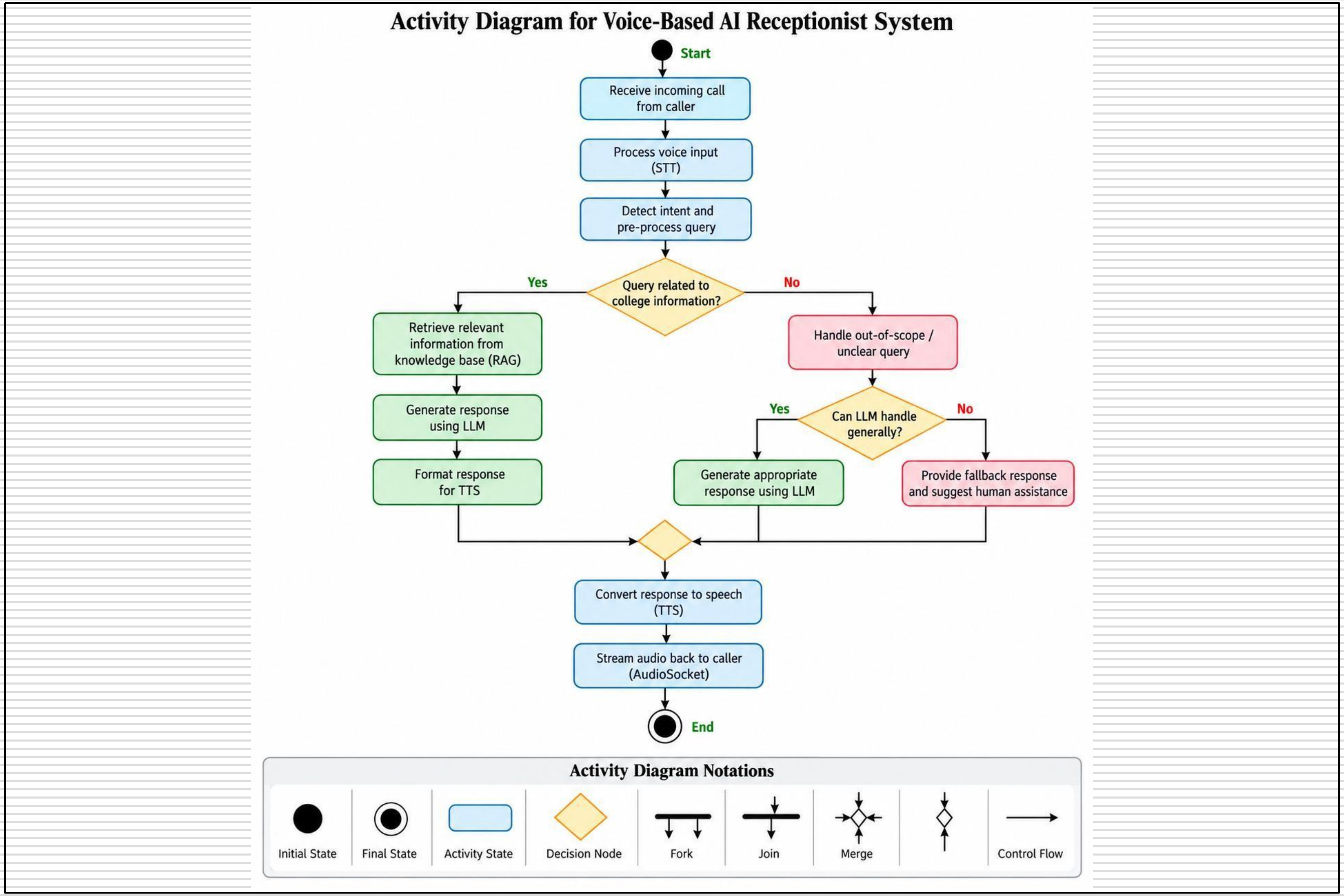
- Official REC Partial webpages knowledge
- REC 2023 UG Regulations
- R2023 CSE Curriculum & Syllabus
- Certificate for Medium of Instruction
- Certificate for CGPA to Percentage Conversion

KNOWLEDGE PROCESSING

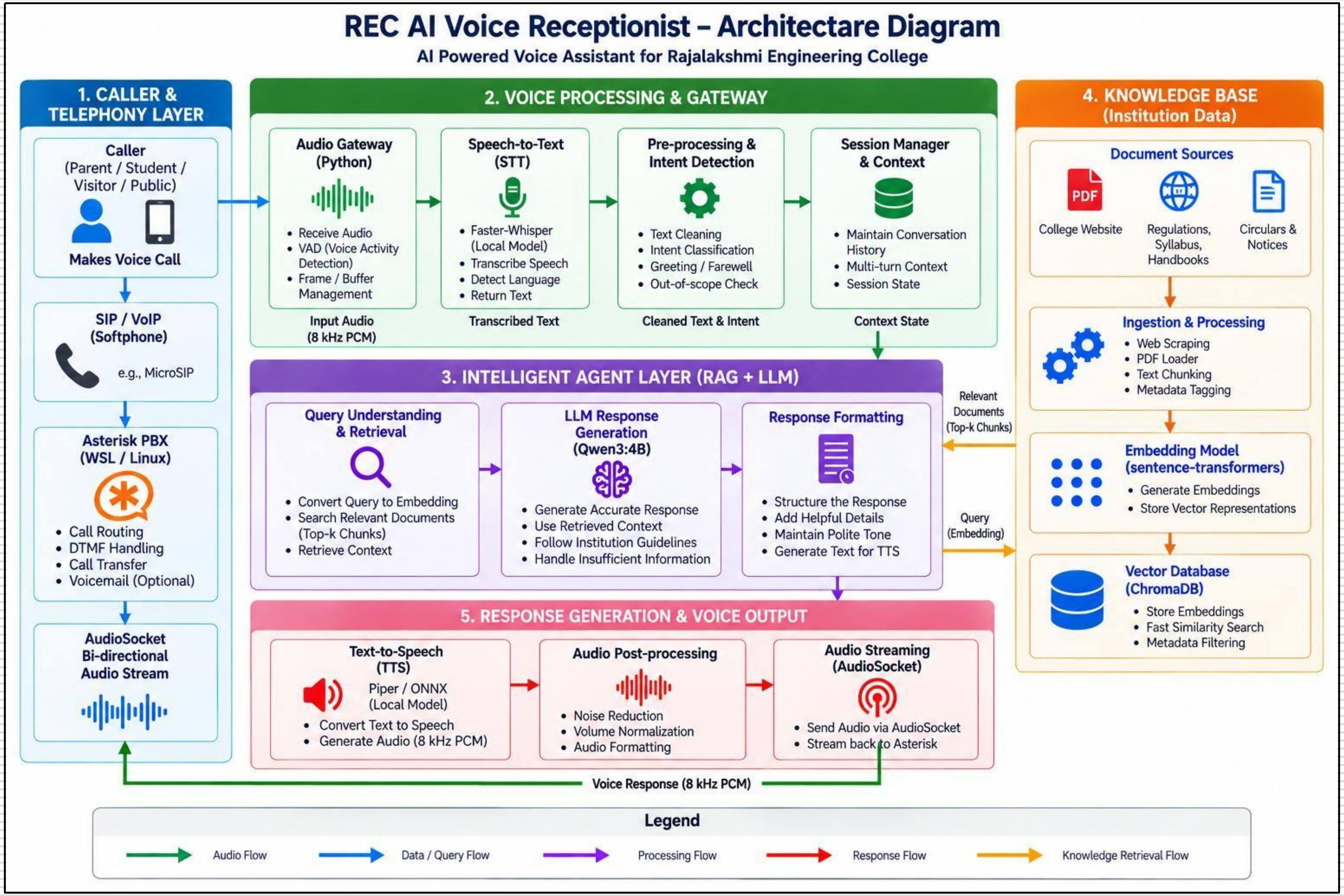
Web/PDF Sources → Text Extraction/OCR → Chunking → Embeddings → ChromaDB

- **767** indexed chunks
- **Sentence-Transformers embeddings**
- **ChromaDB vector store**

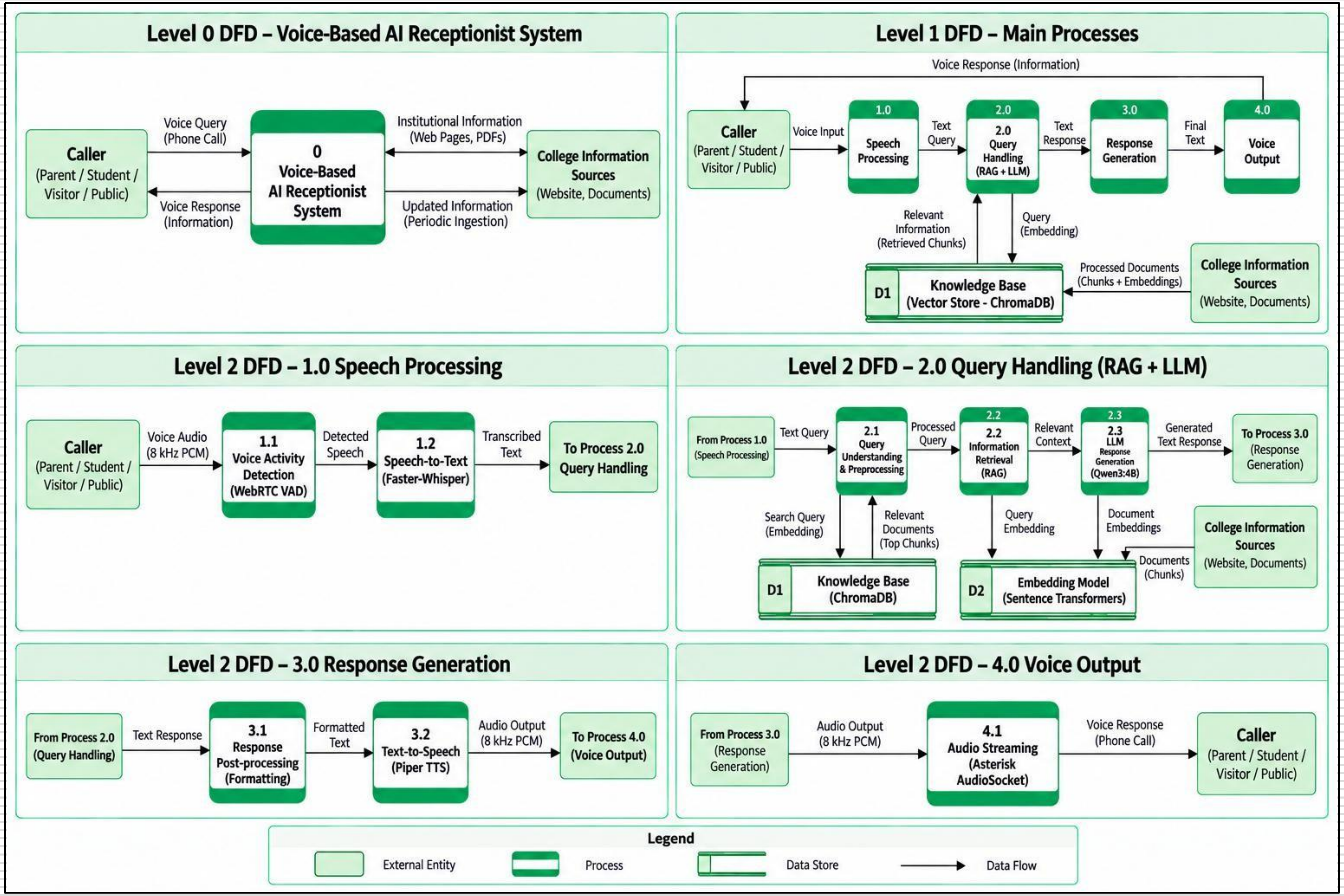
ACTIVITY DIAGRAM



SYSTEM ARCHITECTURE DIAGRAM

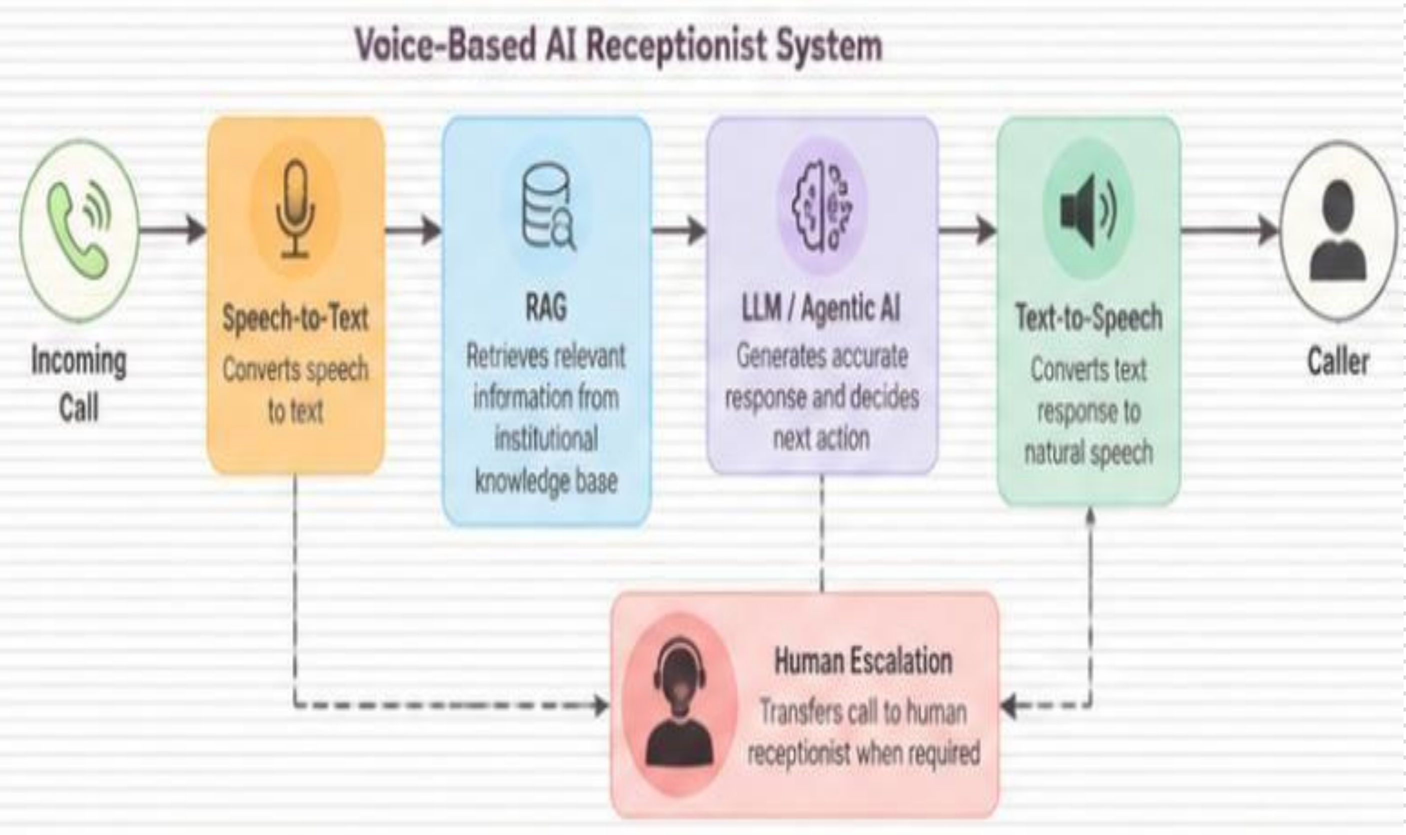


DATA FLOW DIAGRAM



NOVELTY AND KEY INNOVATIONS

USP: A college-specific Voice AI Receptionist that integrates real-time telephony, speech recognition, RAG-based institutional knowledge retrieval, local LLM response generation, and text-to-speech in a single conversational pipeline.



1. **Real-Time Voice Telephony Integration:** Connects **MicroSIP → Asterisk → AudioSocket → Python** to enable live bidirectional voice interaction.
2. **Institution-Specific RAG Knowledge Base:** Uses official **college webpages, academic documents, regulations, and scanned certificates** as the knowledge source for answering institutional enquiries.
3. **OCR-Enhanced Knowledge Processing:** Uses **Tesseract OCR** to extract information from scanned PDF documents before chunking and indexing.
4. **Local AI Processing Pipeline:** Uses **Faster-Whisper, Sentence-Transformers, ChromaDB, Ollama/Qwen3:4B, and Piper TTS** without depending on external AI APIs for the core AI pipeline.
5. **Grounded Response Generation:** The LLM receives retrieved institutional context before generating the answer, helping keep responses relevant to the available college information.
6. **Safe Handling of Unsupported Queries:** When the required information is not available in the knowledge base, the system **refuses to answer instead of fabricating information**.

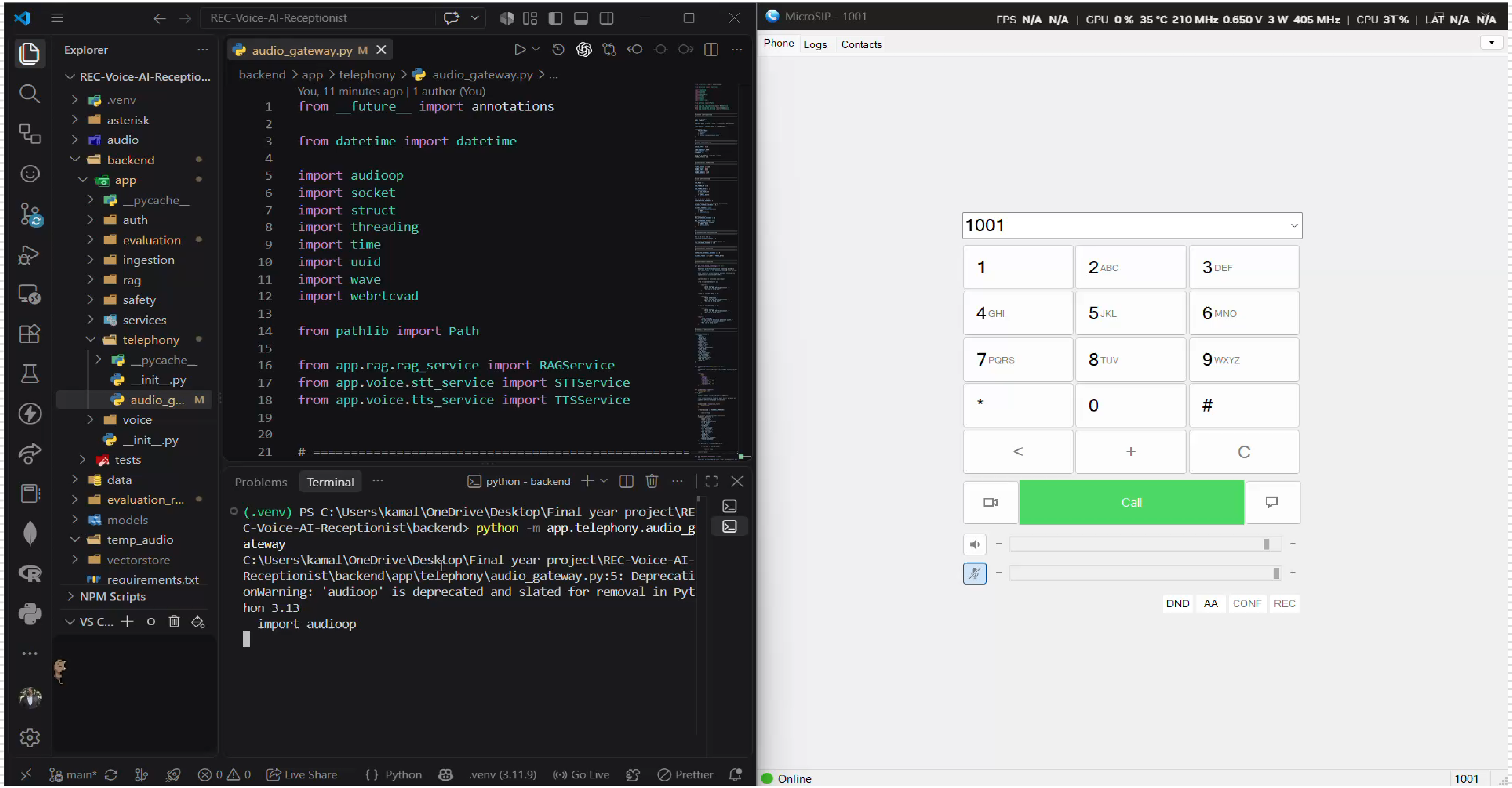
BUSINESS MODEL:

AI Reception-as-a-Service: Automates routine college inquiries, reduces receptionist workload, and provides **24×7 accessible information support**.

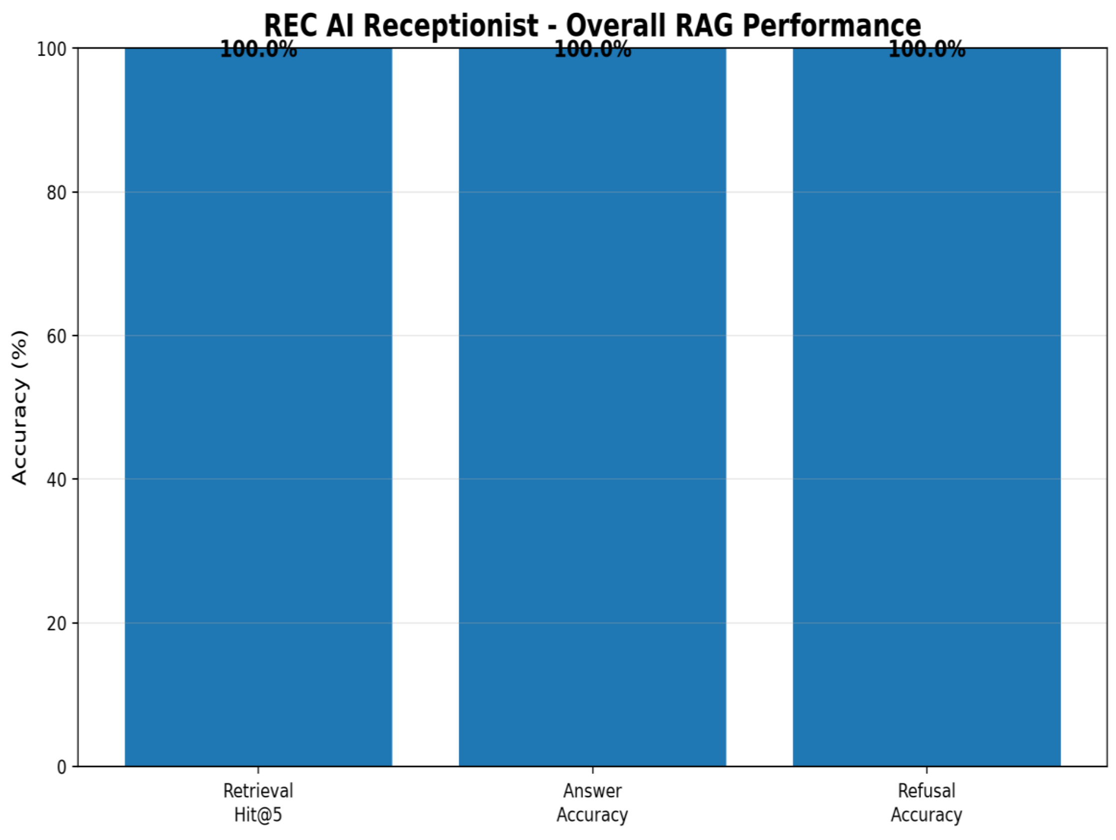
END-TO-END USE CASE

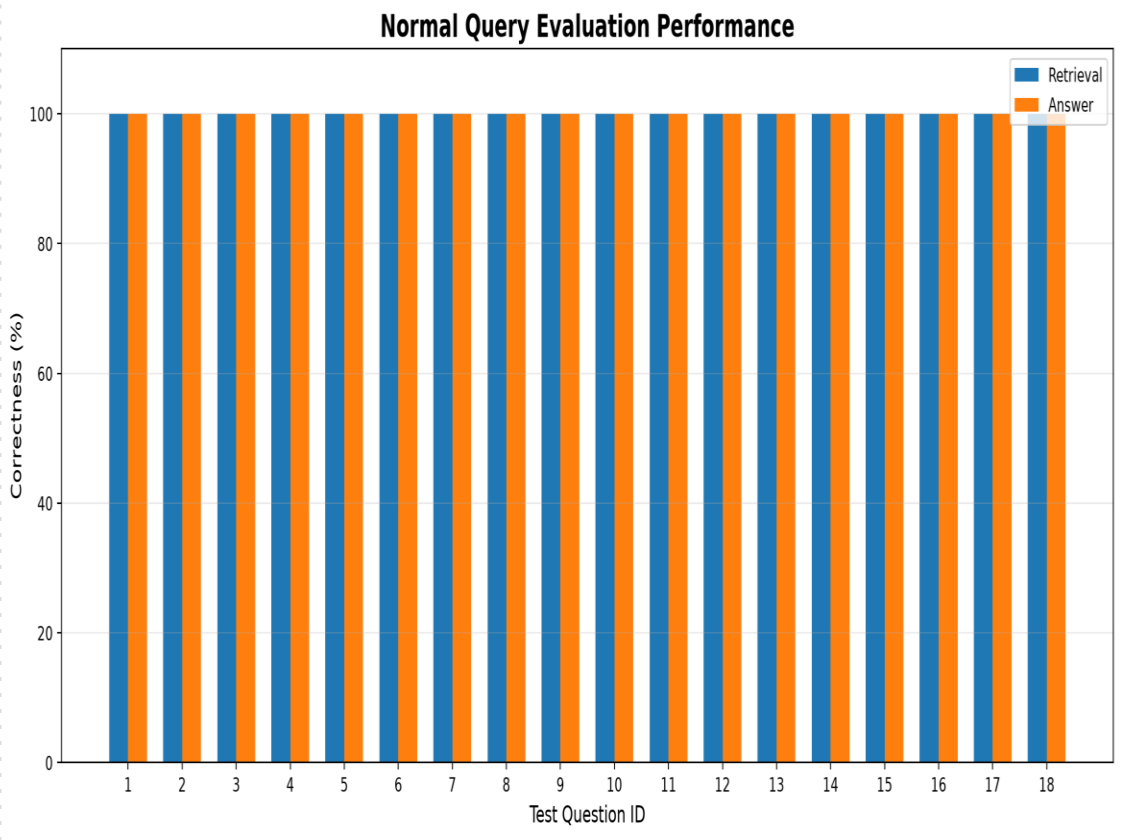
Use Case Step	Module(s) Involved	Status	Function (Notes)
Telephony & Audio	MicroSIP, Asterisk, AudioSocket	**Live**	Handles incoming calls and bidirectional voice communication.
Voice Activity Detection	WebRTC VAD	**Live**	Detects speech start/end and separates speech from silence.
Speech-to-Text (STT)	Faster-Whisper Small, CUDA, FP16	**Live**	Converts caller speech into text queries.
Knowledge Ingestion	PyMuPDF, Tesseract OCR, Web Scraping	**Live**	Extracts and prepares information from college webpages and PDF documents.
RAG Retrieval	Sentence-Transformers, ChromaDB	**Live**	Retrieves relevant institutional information for the caller's query.
LLM Response Generation	Ollama, Qwen3:4B	**Live**	Generates responses using the retrieved institutional context.
Text-to-Speech (TTS)	Piper TTS, ONNX	**Live**	Converts generated responses into natural speech.
Conversation Handling	Session Context, VAD, STT, RAG, LLM, TTS	**Implemented & Tested**	Supports multiple questions and follow-up interaction within a call.

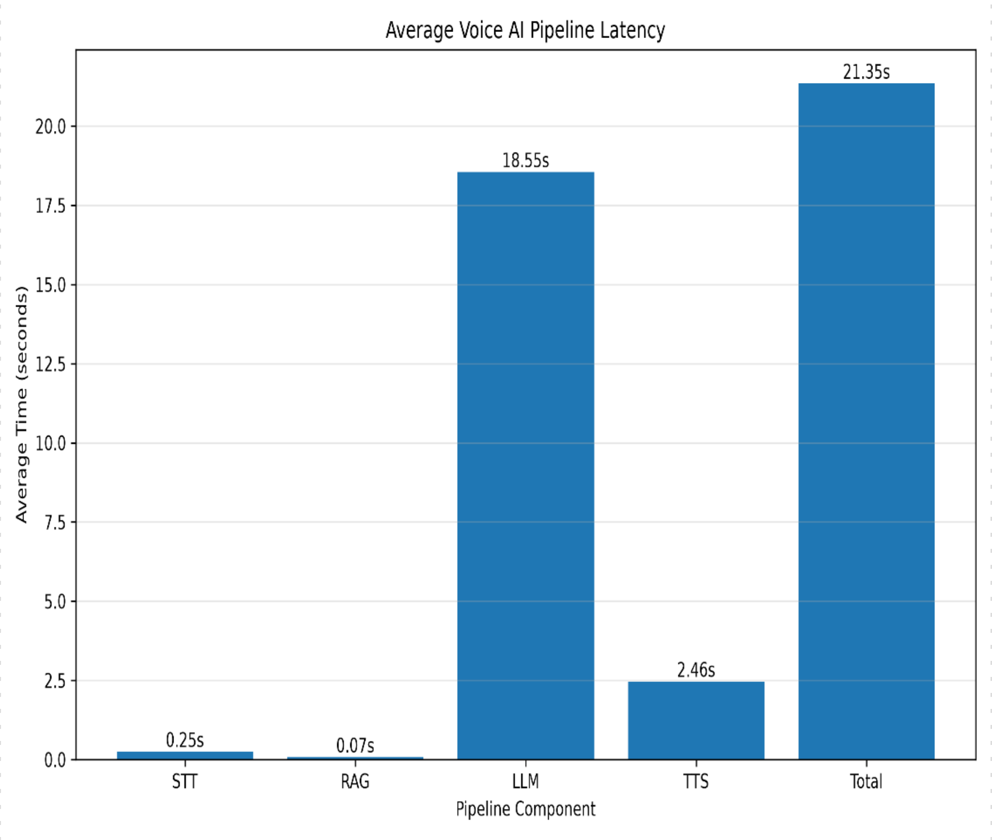
END-TO-END USE CASE – LIVE DEMO



PROJECT RESULT







RAG Evaluation: 20 Test Queries | Retrieval Hit@5: 100% | Answer Accuracy: 100% | Refusal Accuracy: 100%

18/18 normal queries answered correctly and 2/2 unsupported queries correctly refused

The implemented Phase-1 system successfully performs end-to-end voice interaction, retrieves institution-specific knowledge, generates grounded responses, and safely refuses unsupported queries.

CONCLUSION & FUTURE ENHANCEMENTS

PHASE 1 ACHIEVEMENT

- Developed an end-to-end **Voice AI Receptionist** for college information enquiries.
- Integrated **MicroSIP, Asterisk and AudioSocket** for real-time telephone communication.
- Implemented **WebRTC VAD and Faster-Whisper** for speech detection and speech-to-text conversion.
- Built an institution-specific knowledge base from **official college webpages and academic PDF documents**.
- Used **Tesseract OCR** to extract information from scanned documents.
- Implemented **RAG with Sentence-Transformers and ChromaDB** for relevant knowledge retrieval.
- Used **Ollama with Qwen3:4B** to generate context-grounded responses.
- Implemented **Piper TTS** to return responses as voice.
- Supports **multi-turn conversations** and safely refuses unsupported information.

PHASE 2 – PLANNED

Authentication: Secure identification of students before accessing personal information.

- **Student Services Integration:** Connect with **DigiCampus / institutional APIs**.
- **Personalized Queries:** Provide authenticated information such as **attendance, marks, fees and placement eligibility**.
- **Agentic AI:** Introduce tool-based actions and decision-making for complex requests.
- **Human Escalation:** Transfer unresolved or human-required queries to the appropriate receptionist/department.
- **Advanced Language Support:** Improve **Tamil-English code-mixed** and multilingual speech recognition.
- **Performance Optimization:** Reduce LLM processing latency for faster caller response.
- **Knowledge Expansion:** Extend the knowledge base with additional verified institutional sources.

Phase 1 establishes the core voice-based institutional information system, while Phase 2 extends it toward secure, personalized, and intelligent student services.

LITERATURE SURVEY

Author(s) / Year	Title of the Paper	Methodology / Approach	Inference
C. Rzepka, B. Berger & T. Hess (2022)	**Voice Assistant vs. Chatbot – Examining the Fit Between Conversational Agents' Interaction Modalities and Information Search Tasks**	Voice assistant and chatbot interaction modalities Examines information-search tasks Compares suitability of voice and text interaction	Demonstrates that voice-based interaction can be effective for information-search tasks, supporting the use of a voice interface for the proposed receptionist.
A. Radford et al. (2023)	**Robust Speech Recognition via Large-Scale Weak Supervision**	Whisper encoder-decoder Transformer Trained using large-scale multilingual and multitask speech data Robust speech recognition across varied conditions	Demonstrates robust speech recognition across accents, languages, background noise, and different speech conditions, making Whisper suitable for the STT layer of voice applications.
J. Ao et al. (2022)	**SpeechT5: Unified-Modal Encoder-Decoder Pre-Training for Spoken Language Processing**	Unified-modal encoder-decoder architecture Pre-training for spoken language processing Supports multiple speech and text tasks	Shows the effectiveness of unified speech-text processing, supporting the integration of speech processing components in conversational voice systems.

LITERATURE SURVEY

Author(s) / Year	Title of the Paper	Methodology / Approach	Inference
Y. Ren et al. (2021)	**FastSpeech 2: Fast and High-Quality End-to-End Text to Speech**	Non-autoregressive TTS architecture Uses speech-related acoustic representations Designed for fast and high-quality speech synthesis	Demonstrates that efficient TTS can generate high-quality speech with reduced synthesis complexity, supporting real-time voice response generation.
E. Casanova et al. (2022)	**YourTTS: Towards Zero-Shot Multi-Speaker TTS and Zero-Shot Voice Conversion for Everyone**	Zero-shot multi-speaker TTS Zero-shot voice conversion Neural speech synthesis	Shows the capability of neural TTS systems to generate natural speech across different speakers, supporting the use of speech synthesis for conversational voice applications.
S. Borgeaud et al. (2022)	**Improving Language Models by Retrieving from Trillions of Tokens**	Retrieval-enhanced language modelling Retrieves information from a large external corpus Combines retrieval with language-model generation	Demonstrates that retrieving external information can improve language-model performance, providing the foundation for retrieval-based knowledge integration.

LITERATURE SURVEY

Author(s) / Year	Title of the Paper	Methodology / Approach	Inference
S. Siriwardhana et al. (2023)	**Improving the Domain Adaptation of Retrieval Augmented Generation (RAG) Models for Open Domain Question Answering**	Retrieval-Augmented Generation Domain adaptation of RAG models Combines retrieved knowledge with language generation	Demonstrates the importance of domain adaptation in RAG systems, supporting the use of institution-specific knowledge for college-related enquiries.
Y. Gao et al. (2023)	**Retrieval-Augmented Generation for Large Language Models: A Survey**	Reviews retrieval-augmented generation methods Covers knowledge retrieval and generation Analyses RAG architectures and applications	Provides a comprehensive foundation for understanding RAG-based systems and supports the retrieval → context → generation pipeline used in the proposed system.
O. Ram et al. (2023)	In-Context Retrieval-Augmented Language Models	Retrieval-augmented language modelling Retrieved information supplied as context Uses contextual information during generation	Shows how retrieved information can be incorporated directly into the language-model context, supporting context-grounded response generation.

LITERATURE SURVEY

Author(s) / Year	Title of the Paper	Methodology / Approach	Inference
C. Niu et al. (2024)	**RAGTruth: A Hallucination Corpus for Developing Trustworthy Retrieval-Augmented Language Models**	Hallucination corpus for RAG systems Studies unsupported/generated content Focuses on trustworthy retrieval-augmented generation	Highlights the importance of detecting and reducing hallucinations in RAG systems, supporting the need for grounded and reliable responses.
A. Asai et al. (2024)	**Self-RAG: Learning to Retrieve, Generate, and Critique through Self-Reflection**	Retrieval and generation framework Self-reflection and critique Evaluates generated responses during processing	Demonstrates how retrieval and self-critique can improve factuality and generation quality, providing a direction for future improvement of the proposed RAG system.
S. Jeong et al. (2024)	**Adaptive-RAG: Learning to Adapt Retrieval-Augmented Large Language Models through Question Complexity**	Adaptive RAG architecture Considers question complexity Dynamically adapts retrieval and generation	Shows that retrieval strategies can be adapted according to query complexity, supporting future intelligent query routing and handling.

LITERATURE SURVEY

Author(s) / Year	Title of the Paper	Methodology / Approach	Inference
S. Es, J. James, L. Espinosa Anke & S. Schockaert (2024)	**RAGAs: Automated Evaluation of Retrieval Augmented Generation**	Automated RAG evaluation framework Evaluates retrieval and generated responses Uses multiple evaluation metrics	Provides an automated approach for evaluating RAG systems, supporting systematic measurement of retrieval and answer quality in the proposed system.
J. Saad-Falcon, O. Khattab, C. Potts & M. Zaharia (2024)	**ARES: An Automated Evaluation Framework for Retrieval-Augmented Generation Systems**	Automated evaluation framework for RAG Evaluates context relevance Evaluates answer faithfulness and answer relevance	Demonstrates systematic evaluation of RAG quality and provides a strong basis for measuring the reliability of retrieval and generated responses.
A. Mahmood, J. Wang, B. Yao, D. Wang & C.-M. Huang (2025)	**User Interaction Patterns and Breakdowns in Conversing with LLM-Powered Voice Assistants**	Studies user interactions with LLM-powered voice assistants Analyses interaction patterns Examines conversational breakdowns	Highlights challenges in intent recognition and conversational interaction, supporting the need for robust multi-turn voice interaction in the proposed receptionist.

Thank You